12TH BIOLOGY · CHAPTER 19 · FULL CHAPTER NOTES

Chromosome and DNA

Topic-wise short and long questions with answers from the complete chapter

19.1 DETAILED STRUCTURE OF CHROMOSOME

Q.1 [Short] What are chromosomes? How is chromatin different from a chromosome?

Chromosomes are thin, thread-like structures made of DNA and histone proteins. The number and shape of chromosomes are different in each species; this complete set is called the **karyotype**. In the body cells of diploid organisms, chromosomes occur in pairs called **homologous chromosomes**. When a cell is not dividing (interphase), the chromosomes remain uncoiled and form a fine network called **chromatin**. When the cell prepares to divide, the DNA coils around histones and condenses into thick, short chromosomes.

Q.2 [Short] Who discovered chromosomes and who named them?

Chromosomes were discovered by the German embryologist Walther Flemming in 1882, in the rapidly dividing cells of salamander larvae. The name “chromosome” was proposed by the German anatomist Waldeyer-Hartz in 1888. It literally means “coloured bodies”.

Q.3 [Short] What are karyotype and homologous chromosomes?

The complete set of chromosomes of a species, with its specific number and shapes, is called the **karyotype**. In the somatic cells of diploid organisms, chromosomes occur in pairs; these pairs are called **homologous chromosomes**. Gametes have half the number of chromosomes, and they are not in pairs.

Q.4 [Long] Describe the morphology of a duplicated and condensed chromosome.

- **Chromatids:** After duplication in interphase, each chromosome consists of two identical strands called sister chromatids.
- **Centromere:** The constricted region where the sister chromatids are attached. It is a specific DNA sequence of about 220 nucleotides.
- **Kinetochore:** A disc-shaped protein structure attached to the centromere. Here the chromosome attaches to the mitotic spindle during cell division.
- **Secondary constriction** (Nuclear Organizer Region, NOR): present in some chromosomes; it makes nucleoli during interphase.
- **Satellite:** The knob-like end of the chromosome beside the secondary constriction. It has highly repetitive DNA sequences called junk DNA.
- **Telomeres:** The terminal ends of chromosomes. They prevent two chromosomes from attaching with each other at their ends.

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Q.5 [Short] Differentiate between centromere and kinetochore.

The **centromere** is the constricted region of DNA (about 220 nucleotides) where the sister chromatids are attached. The **kinetochore** is a disc-shaped protein structure attached to the centromere; it is the point where the chromosome attaches to the mitotic spindle.

Q.6 [Short] What is the function of telomeres?

Telomeres are the terminal ends of chromosomes. They prevent two chromosomes from attaching with each other at their ends.

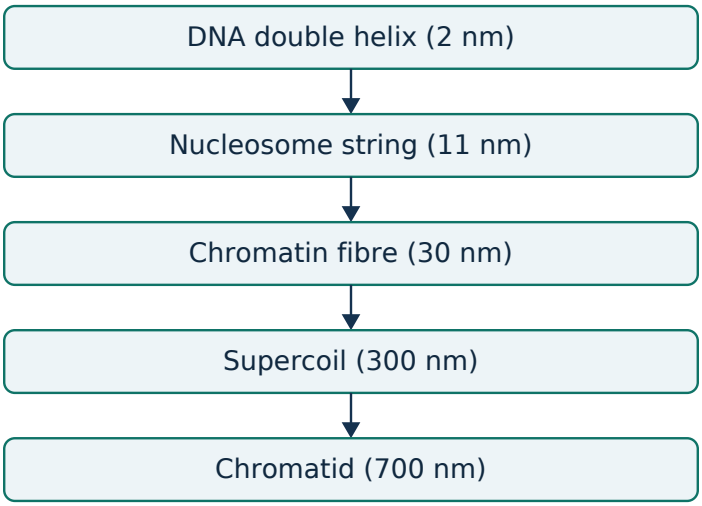
Q.7 [Short] What are the four types of chromosomes on the basis of position of centromere?

Type of chromosome	Position of centromere
Metacentric	At the centre
Sub-metacentric	Slightly away from the centre
Acrocentric	Near the end
Telocentric	At the end

Q.8 [Long] Describe the chemical composition and the levels of organization of a chromosome.

A chromosome is made up of 40% DNA and 60% proteins (histones). The DNA of an average human chromosome is about 5 cm long and has about 140 million nucleotides. DNA is negatively charged due to its phosphate groups. It has a strong affinity for histones, which are positively charged due to basic amino acids such as arginine and lysine. The levels of organization are:

1. **Nucleosome string (11 nm):** About every 146 nucleotides of DNA (2 nm thick) wrap twice around a **histone octamer** (two copies each of H2A, H2B, H3 and H4). This forms a **nucleosome**. Nucleosomes are connected by linker DNA of about 80 nucleotides. The fifth histone, H1 (linker histone), lies at the base of the nucleosome and helps in winding the chain. The nucleosomes look like beads on a string.
2. **Chromatin fibre (30 nm):** In the early stages of cell division, the nucleosome string coils about its axis. The tightly packed nucleosomes make a thicker fibre called the chromatin fibre.
3. **Chromatid (700 nm):** The chromatin fibre folds upon itself to make a supercoil (300 nm thick). The supercoil coils and condenses further and takes the shape of a chromatid.



Levels of organization of a chromosome

Q.9 [Short] Why does DNA have a strong affinity for histones?

DNA is negatively charged because of its phosphate groups. Histones are positively charged because they have many basic amino acids such as arginine and lysine. Opposite charges attract, so DNA binds strongly to histones.

Q.10 [Short] What is a histone octamer? What is the role of histone H1?

A histone octamer is a set of 8 histone molecules: two copies each of H2A, H2B, H3 and H4. DNA wraps twice around it to form a nucleosome. The fifth histone, H1 (linker histone), is present at the base of the nucleosome near the DNA entry and exit points; it helps in winding the chain of nucleosomes.

Q.11 [Short] What is chromatin fibre? How is it formed?

In the early stages of cell division, the nucleosome string (11 nm) coils again about its axis. The tightly packed nucleosomes make a thicker fibre of 30 nm called the chromatin fibre.

Q.12 [Short] Differentiate between heterochromatin and euchromatin.

The inactive portions of chromatin produce dark bands and are called **heterochromatin**. The active portions are called **euchromatin**.

Q.13 [Long] Explain the concept of genes, loci and alleles.

In 1909, the Danish botanist Wilhelm Johannsen introduced the term “gene” for the basic unit of heredity. A **gene** is a short segment of DNA which encodes the sequence of amino acids of a particular polypeptide. The whole DNA molecule of a chromosome does not carry messages for proteins; only short segments (genes) do.

The location where a gene is present on a chromosome is called its **locus** (plural: loci). The alternative forms of a gene are called **alleles**. On homologous chromosomes, alleles are present at the same locus. For example, if a chromosome has an allele for eye colour at a particular locus, its homologous chromosome will carry the allele of the same character at the same locus.

19.2 DNA AS THE HEREDITARY MATERIAL

Q.14 [Short] What characteristics must a molecule have to serve as genetic material?

- It must be able to code for the sequence of amino acids in proteins.
- It must be able to replicate itself before cell division.
- It must be in the nucleus of eukaryotic cells.
- It must be able to change over time to account for evolution.

Only DNA fulfils all these requirements.

Q.15 [Long] Describe Griffith's experiment. What did he conclude?

In 1928, the British biologist Fredrick Griffith worked on the pathogenic bacterium *Streptococcus pneumoniae*. It has two strains: the **S-type** has a polysaccharide coat and forms smooth colonies; the **R-type** lacks the coat and forms rough colonies.

Injection given to mice	Result
Live S-type bacteria	Mice developed pneumonia and died (S-type is virulent)
Live R-type bacteria	Mice showed no effects
Heat-killed S-type bacteria	Mice remained healthy
Mixture of heat-killed S-type and live R-type	Mice developed pneumonia and died; live S-type bacteria were found in their blood

Conclusion: The information for making the polysaccharide coat and for virulence had passed from the heat-killed S-type bacteria to the live R-type bacteria. So the R-type made a coat and changed into virulent S-type. Griffith called this process **transformation**, i.e., the transfer of genetic material from one organism to another to change the genetic makeup of the recipient. He could not find the chemical nature (DNA or protein) of the transferred material.

Q.16 [Short] What is transformation?

Transformation is the transfer of genetic material from one organism to another, which changes the genetic makeup of the recipient organism. Griffith observed it when live R-type bacteria changed into virulent S-type bacteria.

Q.17 [Short] Why were the mice killed by a mixture of heat-killed S-type and live R-type bacteria, when neither alone was harmful?

The information for making the polysaccharide coat passed from the dead S-type bacteria into the live R-type bacteria. The R-type bacteria made the coat, which protects them from the host's immune system, and so became virulent S-type bacteria.

Q.18 [Short] How did Avery, MacLeod and McCarty identify the transforming principle?

In 1944, Avery, MacLeod and McCarty discovered the transforming principle. When they removed proteins from dead S-type bacteria and mixed them with live R-type, the mice still developed the disease, so proteins are not responsible for transformation. When they removed DNA from the dead S-type, no transformation occurred and the mice remained healthy. They concluded that "DNA is the fundamental unit of the transforming principle".

Q.19 [Long] Describe the Hershey–Chase experiment.

In 1952, two American biologists, Alfred Hershey and Martha Chase, experimented with **bacteriophages** (viruses that attack bacteria) to confirm that DNA is the genetic material. A bacteriophage has a DNA core and a protein coat. When it infects a bacterium, it injects some hereditary material which directs the production of new phages. The question was whether this material is DNA or protein.

1. Some phages were grown on a medium with radioactive sulphur (^{35}S). It was incorporated into the amino acids of their **protein coats**.
2. Other phages were grown on a medium with radioactive phosphorus (^{32}P). It was incorporated into the phosphate groups of their **DNA**.
3. **Setup 1:** phages with ^{35}S-labelled protein were allowed to infect bacteria. **Setup 2:** phages with ^{32}P-labelled DNA were allowed to infect bacteria.
4. After some time, the infected bacterial cells and the medium were separated by centrifugation and tested for radioactivity.

Setup	Radioactivity found in
1 (^{35}S-labelled protein)	The medium, not in the bacterial cells
2 (^{32}P-labelled DNA)	The bacterial cells, not in the medium

Conclusion: During infection, the DNA of the phage entered the bacterial cells and directed the making of new phages, while the protein coat remained outside. So DNA, not protein, is the hereditary material.

Q.20 [Short] Why did Hershey and Chase use ^{35}S and ^{32}P?

The isotopes served as tracers. Sulphur is present in the amino acids of protein but not in DNA, so ^{35}S labels only the protein coat. Phosphorus is present in the phosphate groups of DNA but not in the protein coat, so ^{32}P labels only the DNA. In this way the two molecules could be followed separately.

19.3 DNA REPLICATION

Q.21 [Short] What is DNA replication? When does it occur?

DNA replication is the process in which a DNA molecule makes its identical copy (replica). As DNA is a double helix, it makes copies of both of its strands. It happens before cell division, during the S-phase of interphase.

Q.22 [Long] Describe the three models proposed for the mechanism of DNA replication.

1. **Conservative model:** The double helix of the parent DNA remains intact and generates a completely new double helix. So one daughter DNA contains both strands of the parent DNA and the other contains both new strands.
2. **Semi-conservative model:** The two strands of the parent DNA separate. Each strand acts as a template for the synthesis of a new strand. So both daughter DNAs consist of one parent strand and one new strand.
3. **Dispersive model:** The parent double helix is broken and dispersed into nucleotides, which combine with new nucleotides. So both daughter DNAs contain strands which are mixtures of old and new nucleotides.

Q.23 [Long] Describe the Meselson–Stahl experiment.

In 1958, two American biologists, Mathew Meselson and Franklin Stahl, tested the three models and proved that DNA replication is semi-conservative.

1. They grew bacteria for many generations in a medium containing the heavy isotope of nitrogen (^{15}N). The DNA of these bacteria became denser than normal.
2. They transferred these bacteria to a medium containing normal ^{14}N, so that any new DNA would be made with ^{14}N.
3. They collected DNA at different intervals, dissolved it in cesium chloride solution and carried out ultracentrifugation. Each DNA settled where its density matched the density of the cesium; denser DNA moved further down.

Sample	Sediment(s) observed	Explanation
0 minutes (before replication)	One band of heavy density	Both strands of all DNA contain ^{15}N
20 minutes (after first replication)	One band of intermediate density	Each daughter DNA is a hybrid: one ^{15}N strand and one ^{14}N strand
40 minutes (after second replication)	Two bands: one intermediate and one light	Each hybrid DNA gave one hybrid DNA and one light DNA (both strands ^{14}N)

Conclusion: DNA replicates in a semi-conservative manner.

Q.24 [Short] Why did Meselson and Stahl continue the experiment up to the second replication?

After the first replication, the single intermediate band rejected only the conservative model. The semi-conservative and dispersive models both predict intermediate DNA at that stage. After the second replication, two bands (intermediate and light) appeared, which only the semi-conservative model predicts; the dispersive model predicts a single band.

Q.25 [Short] What is a pulse-chase experiment?

An experiment in which radio-isotope labelled DNA is traced in the progeny of an organism. In the **pulse** stage, the organism is exposed to the radio-isotope for a brief time and incorporates it into its DNA. In the **chase** stage, the labelled DNA is observed over time by measuring its radioactivity.

Q.26 [Long] Describe the events of the process of DNA replication.

The basic mechanism of DNA replication is similar in all organisms. The following steps refer to prokaryotes.

1. Opening up the DNA double helix

- Replication begins at a specific sequence of nucleotides called the **replication origin site** (one in prokaryotes, many in eukaryotes). An initiator protein binds to it.
- **Helicases** break the hydrogen bonds between the base pairs. The strands separate and make two Y-shaped **replication forks** (together called a replication bubble).
- **Gyrase** (topoisomerase) attaches ahead of the replication fork and prevents supercoiling.
- **Single-strand binding proteins** bind to the single strands and prevent them from re-joining. Both strands act as templates.

2. Building a primer

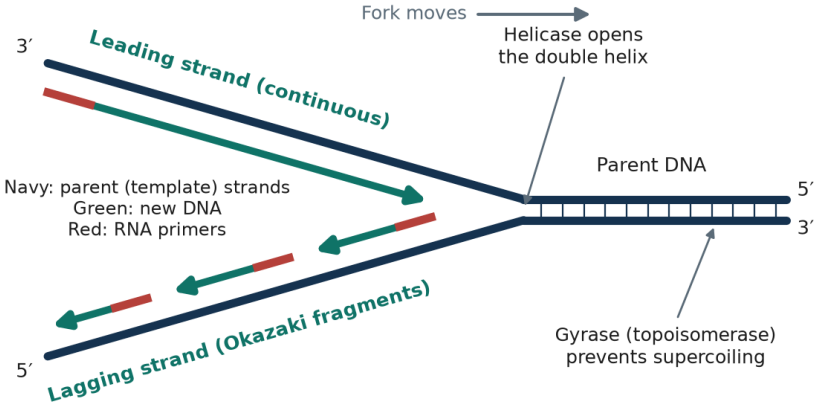
New DNA cannot be started on a bare template. So the enzyme **primase** attaches a short fragment of 5–10 RNA nucleotides, called the **primer**, on each template strand.

3. Assembling complementary strands (extension)

DNA polymerase III is the main enzyme. It matches free nucleotides to the template (A with T, C with G). The raw materials are deoxyribonucleoside triphosphates (dATP, dCTP, dGTP, dTTP); two phosphates break off and release the energy for building. The enzyme can add nucleotides only to the 3′ end, so the new strand grows in the 5′ to 3′ direction.

- **Leading strand:** On the template that runs toward the fork, the enzyme moves in the same direction as the unzipping and builds one continuous strand.
- **Lagging strand:** On the other template, the enzyme must move away from the fork. It has to jump back to a new primer again and again, so the strand is made discontinuously as short pieces called **Okazaki fragments**.

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Replication fork: leading and lagging strands

4. Termination

Replication ends at a specific sequence called the termination site. The region between the origin site and the termination site is called a replication unit or **replicon**. **DNA polymerase I** removes the RNA primers and fills the gaps with DNA nucleotides. **DNA ligase** joins the Okazaki fragments of the lagging strand.

Q.27 [Short] What are replication origin site, replication fork and replicon?

The **replication origin site** is a specific sequence of nucleotides where replication begins. The Y-shaped structure made when the two strands separate is a **replication fork**; two forks together make a replication bubble. The region of DNA between the origin site and the termination site is a replication unit or **replicon**.

Q.28 [Short] State the roles of helicase, gyrase and single-strand binding proteins.

Protein	Role
Helicase	Breaks the hydrogen bonds between the base pairs and unwinds the helix
Gyrase (topoisomerase)	Attaches ahead of the fork and prevents supercoiling
Single-strand binding proteins	Keep the separated strands from re-joining and rewinding

Q.29 [Short] What is a primer? Why is an RNA primer used rather than DNA?

A primer is a short fragment of 5–10 RNA nucleotides which primase attaches to the template, because new DNA cannot be started on a bare template. Starting a chain introduces many errors; RNA marks this initial stretch as “temporary”, so the error-prone part is easily cut out later.

Q.30 [Short] State the functions of DNA polymerase I, II and III.

Polymerase III is the main enzyme: it adds new nucleotides along both template strands and proofreads. **Polymerase I** supports it by removing the primers at the end of replication and filling the gaps. **Polymerase II** repairs DNA damage during the lifetime of a cell.

Q.31 [Short] From where does the energy for DNA synthesis come?

The new nucleotides are added as deoxyribonucleoside triphosphates (dATP, dCTP, dGTP, dTTP). As DNA polymerase III attaches a nucleotide, two phosphates break off. This releases the energy used for building the strand.

Q.32 [Short] Why are the leading and lagging strands synthesised differently?

DNA polymerase III can add nucleotides only to the 3′ end, so it builds only in the 5′ to 3′ direction. But the two template strands run in opposite directions. So on one template the enzyme moves with the fork and builds continuously (leading strand), while on the other it moves away from the fork and builds short Okazaki fragments (lagging strand).

Q.33 [Short] What are Okazaki fragments? How are they joined?

They are the short, separate pieces of DNA made on the lagging strand, named after the Japanese scientist who discovered them. Each needs its own primer. DNA polymerase I replaces the primers with DNA, and DNA ligase joins the fragments.

Q.34 [Long] Differentiate between DNA replication in prokaryotes and eukaryotes.

Feature	Prokaryotes	Eukaryotes
Location	Cytoplasm	Nucleus
Polymerases	DNA polymerase I, II and III	At least 14 polymerases (e.g., alpha, delta, epsilon)
Origin and termination sites	Single origin and single termination site	Multiple sites on each chromosome; several replicons
Okazaki fragments	Longer (1000 to 2000 nucleotides)	Short (100 to 200 nucleotides)
Speed	Small DNA is replicated very quickly	Massive linear DNA; replication starts at thousands of origin sites at the same time

Q.35 [Short] How do biotechnologists use DNA replication?

- **PCR:** to make billions of copies of a specific DNA.
- **Recombinant DNA technology:** to amplify DNA cut from different sources.
- **Genome sequencing:** to make multiple copies of the genome for sequencing.
- **Diagnostic studies:** to replicate sections of a patient's DNA and search them for mutations.

Q.36 [Short] Describe stability and variability as two properties of replicating DNA.

Stability means the ability of DNA to be copied with accuracy. DNA polymerase III has a proofreading ability, which ensures that the sequence of nucleotides is maintained. This maintains the specific traits of an organism and the correct functioning of cells.

Variability is the exception. Occasionally a permanent change in the DNA sequence occurs, known as a mutation. It may be caused by errors during replication or by environmental factors. Variability is the primary source of new traits; it allows populations to adapt and is the raw material for evolution.

19.4 GENE EXPRESSION AND GENETIC CODE

Q.37 [Short] State the central dogma of gene expression.

The English biologist Francis Crick used the term “Central Dogma” for the basic mechanism of gene expression. It states that information passes from the genes (DNA) to an RNA copy of the gene, and the RNA copy directs the sequential assembly of a chain of amino acids:

DNA → RNA → Protein

The flow of genetic information is one-way. It consists of two steps: **transcription** (synthesis of RNA from DNA) and **translation** (synthesis of a protein by using the information in RNA).

Q.38 [Long] Describe the work of Crick, Nirenberg, Leder and Khorana on the genetic code.

Francis Crick (1961): He and his colleagues proposed that the genetic code consists of a series of blocks of information called codons, each specifying one amino acid. A codon of 1 nucleotide could code only 4 amino acids, and of 2 nucleotides only 16 (4 × 4). So a codon must be a sequence of 3 nucleotides, which gives 64 (4 × 4 × 4) combinations, more than enough for 20 amino acids. So the genetic code is a triplet code.

Nirenberg, Leder and Khorana: In 1961, Marshall Nirenberg made a poly-U mRNA and added it to the cytoplasm of bacteria. They produced a polypeptide of phenylalanine only, so UUU specifies phenylalanine. Nirenberg and Philip Leder determined the amino acids of 47 codons, and Har Gobind Khorana determined the remaining 17.

Out of 64 codons, three (UGA, UAG and UAA) are **stop codons** or nonsense codons; they specify no amino acid and stop protein synthesis. AUG is the **start codon**; it specifies methionine.

Q.39 [Short] Why must a codon consist of three nucleotides?

There are 4 nucleotides and 20 amino acids. One-nucleotide codons would code only 4 amino acids, and two-nucleotide codons only 16 (4 × 4). Three-nucleotide codons give 64 (4 × 4 × 4) combinations, which are more than enough for 20 amino acids.

Q.40 [Short] Differentiate between genetic code and codon.

The **genetic code** is the information in DNA for the synthesis of proteins. It consists of a series of **codons**; each codon is a sequence of 3 nucleotides that specifies one amino acid. The term codon usually refers to the codon of mRNA (e.g., TAC on DNA means AUG on mRNA).

Q.41 [Short] What are start and stop codons?

AUG is the **start codon**; it is present at the beginning of the message and specifies methionine. UGA, UAG and UAA are **stop codons** (nonsense codons); they do not specify any amino acid and stop protein synthesis.

Q.42 [Short] Who received the 1968 Nobel Prize related to the genetic code?

Nirenberg (work on the genetic code), Har Gobind Khorana (synthesis of nucleic acids) and Robert Holley (chemical structure of transfer RNA). Khorana completed his studies at Government College, Lahore.

Q.43 [Long] Describe the characteristics of the genetic code.

1. **Universal:** The genetic code is the same in almost all organisms; e.g., AGA specifies arginine in bacteria, humans and all other organisms. So genes can be transferred among organisms. Exceptions: the code of mitochondrial DNA is slightly different; some ciliates and bacteriophages also use different codes.
2. **Triplet:** The code is a sequence of three nucleotides (codons). So the information in DNA is enough to specify 20 kinds of amino acids.
3. **Degenerate:** Multiple codons can specify the same amino acid. This is important for mutation tolerance and evolutionary flexibility.
4. **No punctuation:** A codon is read continuously, without gaps between its three nucleotides.
5. **Non-overlapping:** Each triplet codon is read independently of the adjacent codons. For example, AUGAGCGCA is read only as AUG/AGC/GCA, not as AUG/UGA/GAG.

For degeneracy, the textbook (p.132) gives the example that AUG, GUG, UUG and CUG all code for methionine. In the standard genetic code only AUG codes for methionine. A safe example from the same chapter (p.144) is GGC and GGU, which both code for glycine.

Q.44 [Short] What are the exceptions to the universality of the genetic code?

The genetic code of mitochondrial DNA is slightly different from that of chromosomal DNA, which reflects the independent evolutionary history of mitochondria. Some organisms (e.g., ciliates) and viruses (e.g., bacteriophages) also use different codes for the same amino acids.

Q.45 [Short] How many types of tRNA are necessary for a cell if the genetic code is a triplet code?

Of the 64 codons, 3 are stop codons with no anticodon, so 61 codons specify amino acids. With one tRNA for each of these codons, 61 types of tRNA are needed; at least 20 are necessary, one for each amino acid.

19.5 TRANSCRIPTION

Q.46 [Long] Explain the mechanism of transcription.

Transcription is the first step of gene expression, in which a complementary RNA is synthesised from DNA. Only one strand of DNA, the **template strand**, is transcribed. The other strand is the **coding strand**; it has the same sequence as the mRNA, except that T takes the place of U.

1. Initiation phase

No primer is needed. The enzyme **RNA polymerase** binds to the template DNA at a special nucleotide sequence called the **promoter**, which is not itself transcribed. In prokaryotes the promoters are TTGACA (–35 sequence) and TATAAT (–10 sequence). In eukaryotes they are TATAAA (TATA box, –25 sequence) and GGCCAATCT (CAAT box, –70 sequence).

RNA polymerase has five subunits: two α subunits (bind regulatory proteins), a β′ subunit (binds template DNA), a β subunit (binds RNA nucleotides) and a σ subunit (recognizes the promoter and initiates synthesis). After the enzyme attaches to the promoter, the σ subunit is removed. The other four subunits (core enzyme) unwind the DNA helix and form the **transcription bubble**.

2. Elongation phase

Inside the transcription bubble, RNA polymerase makes a chain of complementary ribonucleotides along the template strand in the 5′ to 3′ direction, usually starting with A or G. The nucleotides are added as ribonucleoside triphosphates (rNTP); two phosphates detach and release energy. The first 12 bases of the new mRNA make a temporary helix with the template DNA, which keeps the 3′ end of the mRNA at the catalytic site. The bubble moves down the DNA, the growing mRNA protrudes from it, and the transcribed DNA is re-wound behind it.

3. Termination phase

At the end of the gene there is a stop region: a series of GC base pairs followed by a series of AT pairs. When it is transcribed, hydrogen bonds form within the mRNA and make a loop called the **GC hairpin**, followed by a tail of four or more U ribonucleotides. The hairpin makes RNA polymerase pause. The pairing of the U's of mRNA with the A's of DNA is the weakest, so it cannot hold the RNA–DNA hybrid together. The RNA strand dissociates from the DNA and transcription stops.

The textbook (p.135) writes that there is a "stop codon" at the end of a gene on the template DNA before the stop region. Transcription is actually ended by the stop region (GC hairpin), as described above; stop codons act in translation.

Q.47 [Short] Differentiate between template strand and coding strand.

The **template strand** (antisense, –) is the DNA strand that is transcribed; the mRNA is complementary to it. The **coding strand** (sense, +) is not transcribed; it has the same sequence as the mRNA, except that it has T in place of U.

Q.48 [Short] What is a promoter? Give examples.

A promoter is a special nucleotide sequence on DNA where RNA polymerase binds to begin transcription; it is not itself transcribed. Prokaryotes: TTGACA (–35 sequence) and TATAAT (–10 sequence). Eukaryotes: TATAAA (TATA box, –25) and GGCCAATCT (CAAT box, –70).

Q.49 [Short] Name the subunits of RNA polymerase with their functions.

Subunit	Function
Two α	Bind with regulatory proteins
β′	Binds with template DNA
β	Binds with RNA nucleotides
σ	Recognizes the promoter and initiates synthesis; removed afterwards

Q.50 [Short] How do RNA polymerases differ in prokaryotes and eukaryotes?

Prokaryotes have only one RNA polymerase, which makes all three kinds of RNA. Eukaryotes have three: RNA polymerase I makes rRNA in the nucleolus, RNA polymerase II makes mRNA, and RNA polymerase III makes tRNA.

Q.51 [Short] What is the GC hairpin? How does it end transcription?

When the stop region (GC pairs followed by AT pairs) is transcribed, base pairing within the mRNA makes a loop called the GC hairpin, followed by four or more U's. The hairpin makes RNA polymerase pause. The U–A pairing is the weakest and cannot hold the RNA–DNA hybrid, so the RNA separates from the DNA.

Q.52 [Long] Explain the post-transcriptional modifications of mRNA in eukaryotes.

In prokaryotes, transcription occurs in the cytoplasm, so the new mRNA is translated directly. In eukaryotes, the new mRNA has to travel from the nucleus to the cytoplasm. For this journey it undergoes the following modifications:

1. **Addition of 5′ cap:** A methylated GTP is attached to the first nucleotide (A or G) at the 5′ end. This 5′ cap protects the mRNA from nucleases and phosphatases.
2. **Addition of 3′ poly-A tail:** A chain of about 250 A nucleotides is added at the 3′ end. It protects the mRNA from degradation by nucleases.
3. **Splicing:** The new mRNA is immature because it contains **introns** (regions that do not code for proteins) and **exons** (regions that are expressed). In splicing, the introns are removed and the remaining exons are joined. So the mature mRNA consists of coding segments only.

Q.53 [Short] Why does the eukaryotic mRNA become shorter before it enters the cytoplasm?

The immature mRNA contains non-coding introns. During splicing in the nucleus, the introns are removed and only the exons are joined. So the mature mRNA is shorter than the transcribed mRNA.

19.6 TRANSLATION

Q.54 [Long] Describe the mechanism of protein synthesis (translation).

In translation (protein synthesis), the information present in an mRNA transcript is used to direct the sequence of amino acids of a polypeptide.

Activation of amino acids

Specific tRNAs attach with specific amino acids and make aminoacyl-tRNAs (aa-tRNA). The enzyme **aminoacyl-tRNA synthetase** carries out this reaction; there are 20 kinds of it, one for each amino acid. The aa-tRNA delivers the amino acid to the ribosome.

1. Initiation phase

1. The aa-tRNA carrying the first amino acid binds to the smaller ribosomal subunit under the control of an initiation factor. The first amino acid is N-formyl-methionine in prokaryotes and methionine in eukaryotes.
2. Another initiation factor attaches the 5′ end of mRNA to the smaller subunit and positions the anticodon of the aa-tRNA over the first codon (AUG).
3. The larger ribosomal subunit is placed upon the smaller subunit. This forms the **initiation complex**.

The ribosome has three sites: the **P site** (peptidyl site, where peptide bonds form), the **A site** (aminoacyl site, where the next aa-tRNA binds) and the **E site** (exit site, where empty tRNAs leave). They are oriented 5′ to 3′ as E–P–A.

2. Elongation phase

1. The first aa-tRNA occupies the P site. The next aa-tRNA, with a matching anticodon, binds to the A site with the help of an elongation factor.
2. The enzyme **peptidyl transferase** removes methionine from the first tRNA at the P site and attaches it by a peptide bond to the amino acid of the second aa-tRNA at the A site.
3. The ribosome moves three nucleotides along the mRNA in the 5′–3′ direction. This movement is called **translocation**.
4. The tRNA carrying the chain of two amino acids shifts from the A site to the P site, a new codon is exposed at the A site, and the empty tRNA leaves from the E site.
5. These steps are repeated until a stop codon reaches the A site.

3. Termination phase

When a stop codon is exposed at the A site, no aa-tRNA comes there, because there is no anticodon for stop codons. The stop codon is recognized by a protein called the **release factor**. It releases the polypeptide from the tRNA at the P site. The tRNA is released and the ribosomal subunits separate from the mRNA.

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Q.55 [Short] How are amino acids activated before translation?

Specific tRNAs attach with specific amino acids and make aminoacyl-tRNAs. The enzyme aminoacyl-tRNA synthetase carries out this reaction. There are 20 kinds of this enzyme; each binds a specific amino acid and recognizes the tRNA with the anticodons for that amino acid.

Q.56 [Short] What are the P, A and E sites of a ribosome?

P site (peptidyl site): where peptide bonds form. **A site** (aminoacyl site): where the next aa-tRNA binds. **E site** (exit site): where empty tRNAs leave the ribosome. With respect to mRNA they lie 5′ to 3′ as E–P–A, because the ribosome moves toward the 3′ end.

Q.57 [Short] What is translocation?

It is the movement of the ribosome by three nucleotides along the mRNA in the 5′–3′ direction. As a result, the tRNA carrying the growing chain shifts from the A site to the P site and a new codon is exposed at the A site.

Q.58 [Short] Why does no aminoacyl-tRNA bind when a stop codon reaches the A site?

There is no anticodon for the stop codons, so no tRNA matches them. Instead, a protein called the release factor recognizes the stop codon and releases the polypeptide from the tRNA at the P site.

Q.59 [Long] Differentiate between translation in prokaryotes and eukaryotes.

Prokaryotes	Eukaryotes
Genes lack introns.	Most genes possess introns.
After transcription, mRNA is translated immediately; there is no post-transcriptional modification.	mRNA is modified before translation: introns are cut out, a 5′ cap and a 3′ poly-A tail are added.
Translation occurs in the cytoplasm.	Translation occurs in the cytoplasm as well as on the rough endoplasmic reticulum.
Ribosomes are smaller (70S).	Ribosomes are larger (80S).
The initial amino acid is modified, i.e., N-formyl methionine.	The initial amino acid is methionine; it is not modified.

19.7 REGULATING GENE EXPRESSION

Q.60 [Long] What is regulation of gene expression? Describe its importance.

All the genes of an organism are not expressed at the same time, and all genes are not turned ON in all cells. Turning genes ON and OFF and the timing of their expression are controlled by regulator proteins; this is called the **regulation of gene expression**. Its importance:

- It enables cells to respond to environmental conditions and to prepare a protein only when it is needed.
- In multicellular organisms, it is the basis of cellular differentiation and morphogenesis. It creates different cell types: the genome is the same, but different genes are expressed in each type.
- Example: liver cells express the genes for the enzyme alcohol dehydrogenase, while neurons of the brain keep these genes turned OFF.

Q.61 [Short] At which levels is gene expression regulated?

Level	What is regulated
Transcriptional	The rate of transcription
Post-transcriptional	The modification of mRNA
Translational	The rate of translation
Post-translational	The modification of protein (folding, stability etc.)

Q.62 [Long] Describe the negative and positive control of gene expression.

At the transcriptional level, genes have special regulatory sequences to which specific regulatory proteins bind.

Negative gene regulation: Regulatory proteins called **repressors** bind to pieces of DNA called **operators**, present near the promoter. Here the repressors block RNA polymerase and stop or reduce transcription.

Positive gene regulation: Regulatory proteins called **activators** (or inducers) bind to the operators. They start transcription or increase its speed by helping RNA polymerase to bind to the promoter.

Q.63 [Short] How is gene expression regulated during splicing?

In eukaryotes, genes can also be regulated during mRNA splicing. Specific regulatory proteins in specific cell types remove pieces of mRNA (introns or exons) in different ways. So mRNAs of different lengths are produced, and more than one polypeptide is made from a single gene (alternate splicing).

Q.64 [Short] All cells have the same genome. Why do liver cells make alcohol dehydrogenase but neurons do not?

Specific genes are turned ON in each cell type. Liver cells express the genes for alcohol dehydrogenase, which breaks alcohol into non-toxic molecules. Neurons also have these genes but keep them unexpressed (turned OFF).

19.8 MUTATIONS

Q.65 [Short] Define mutation, mutagen and mutant.

A change in the content of the genetic message is called a **mutation**. The agents that cause mutations are called **mutagens**. The organism or cell in which a mutation occurs is called a **mutant**.

Q.66 [Short] Differentiate between natural and induced mutations.

Natural mutations	Induced mutations
Occur spontaneously, due to sequencing errors or DNA damage during replication.	Caused by external factors: mutagen substances, radiation, pesticides, industrial pollutants, chemotherapy.
A part of the process of evolution; effects may be beneficial, neutral or harmful.	Can have serious health effects on the organism.

Q.67 [Short] How do mutations contribute to evolution?

Mutation and recombination are the ultimate sources of all genetic variation. They create new alleles and new combinations of alleles, which form new traits (advantageous, harmful or neutral). Natural selection acts on these variations and the frequencies of advantageous alleles increase. In the long run this leads to evolution.

Q.68 [Short] What is a gene mutation?

It is a general term for any change within the chemical structure of a single gene. It is a change at the molecular level (nucleotide sequence), not at the chromosomal level. Point mutations are the most common gene mutations.

Q.69 [Long] Write a note on point mutations and their sources.

A change in the content of the genetic message is called a mutation. Mutations involving a change in only one or a few nucleotides are called **point mutations**. Common examples:

1. **Substitutions:** exchange of a single nucleotide for another, e.g., A is substituted with G.
2. **Insertions:** addition of one or more extra nucleotides into the DNA, usually due to errors during replication.
3. **Deletions:** removal of one or more nucleotides from the DNA.

Sources of point mutations

1. **Physical damage to DNA:** *Ionizing radiation* (X-rays, gamma rays) changes atoms into free radicals, which may break the phosphodiester bonds of DNA. *Ultraviolet radiation* is strongly absorbed by DNA; a double covalent bond forms between two adjacent pyrimidine nucleotides (C and T), which can block replication.
2. **Mutagen chemicals:** Some resemble nucleotides and are incorporated into DNA; some remove the amino group from adenine or cytosine; some add hydrocarbon

groups to nucleotides. All cause mis-pairing. Example: mustard gas.

3. **Spontaneous point mutations:** Sometimes nucleotides spontaneously shift to alternative positions, and DNA polymerase pairs a different nucleotide with them. Sometimes sequences misalign when homologous chromosomes pair; a loop forms, the cell cuts it out, and several hundred nucleotides are deleted.

Q.70 [Short] How does ultraviolet radiation cause mutation?

DNA strongly absorbs UV radiation. When two adjacent pyrimidine nucleotides (C and T) absorb it, a double covalent bond forms between them. This can completely block replication.

Q.71 [Short] Enlist the commonly occurring minor mutations in humans.

- **Single nucleotide polymorphisms (SNPs):** involve a single nucleotide; very common; harmless or with subtle effects, e.g., inability to digest milk (lactose).
- **Insertions and deletions:** add or remove one or more nucleotides, e.g., cystic fibrosis is commonly caused by a deletion of just three nucleotides.
- **Silent mutations:** do not change the amino acid sequence, e.g., GGC to GGU, both code for glycine.
- **Missense mutations:** change only one amino acid, e.g., glutamic acid replaced by valine in haemoglobin (sickle cell anaemia).
- **Nonsense mutations:** create a premature stop codon, giving a short, non-functional protein, e.g., Duchenne muscular dystrophy.

Q.72 [Short] Justify: most mutations are harmful.

Most mutations disrupt the normal functioning of genes.

- A change of a single amino acid in haemoglobin causes sickle cell anaemia.
- Mutations can disrupt gene regulation and developmental processes and contribute to diseases such as cancer.
- Some mutations create non-functional or unstable proteins, e.g., in phenylketonuria.

Q.73 [Long] Describe the causes, symptoms and treatment of sickle cell anaemia.

Sickle cell anaemia is due to a recessive gene for an abnormal β chain of haemoglobin, present on chromosome 11.

	Sickle cell anaemia
Causes	A substitution-type point mutation: the DNA nucleotide T is substituted by A. The codon of glutamic acid (CTC) changes to CAC, which is for valine. A child who receives both mutant alleles makes abnormal β chains.
Symptoms	The abnormal haemoglobin carries less oxygen and the RBCs become sickle-shaped. Pain, fatigue, shortness of breath, jaundice and an increased risk of infections.

	Sickle cell anaemia
Treatment	Pain management with NSAIDs and opioids; blood transfusions to increase healthy RBCs; bone marrow transplant for a long-term cure; vaccinations and antibiotics to prevent complications.

Q.74 [Long] Describe the causes, symptoms and treatment of phenylketonuria.

Phenylketonuria (PKU) is a genetic metabolic disorder. A normal gene on chromosome 12 produces the liver enzyme phenylalanine hydroxylase (PAH), which converts the amino acid phenylalanine into tyrosine.

	Phenylketonuria
Causes	A point mutation (substitution or deletion of a nucleotide) makes the allele recessive. A child with both recessive alleles does not make PAH properly, so phenylalanine builds up to toxic levels in blood and tissues.
Symptoms	Intellectual disability, seizures, behavioural problems, mental disorder, microcephaly (smaller head size), musty odour in breath, skin and urine; brain damage.
Treatment	Avoid foods high in phenylalanine (meat, dairy products, eggs, some grains); regular monitoring of blood phenylalanine; medicines containing synthetic PAH enzyme; nutritional supplements.

Q.75 [Short] What are the possible ways to treat genetic diseases?

- **Gene therapy:** correcting or replacing the mutated gene with a functional gene.
- **Cell therapy:** replacing non-functional cells with healthy ones, e.g., pancreatic beta cells in type 1 diabetes.
- **Pharmacogenomics:** tailoring drug treatment to the patient's genetic makeup.
- **Genetic counselling:** genetic testing, education on the disease and guidance on family planning.

Q.76 [Short] What are chromosomal mutations? Name their types.

They are large-scale mutations that involve changes in the position of genes, or in the structure or number of chromosomes. They are usually spontaneous. Types: (1) changes in chromosome structure (chromosomal rearrangements and insertional inactivation); (2) changes in chromosome number (chromosomal aberrations) due to non-disjunction.

Q.77 [Short] What are chromosomal rearrangements?

They are physical alterations of chromosomes: deletions, duplications, inversions and translocations. They are usually caused by breakage of the DNA double helix at two different locations. In a translocation, a segment of one chromosome becomes part of another chromosome. In an inversion, the orientation of a portion of a chromosome is reversed.

Q.78 [Short] What are transposons? What is insertional inactivation?

Transposons (jumping genes) are small mobile segments of DNA that move from one location to another in the genome at random. When a transposon is inserted into a gene, it inactivates the gene; this is insertional inactivation. Example: a transposon inserted into the clotting factor IX gene on the X chromosome places a stop codon in the gene and causes X-linked haemophilia.

Q.79 [Short] What is non-disjunction?

Occasionally, sister chromatids fail to separate properly in meiosis, so a gamete gains or loses a chromosome. This is called **non-disjunction**. It can happen in autosomes or in sex chromosomes.

Q.80 [Short] Why is the age of the mother more critical than that of the father for non-disjunction?

All the future eggs of a female have developed up to prophase I by the time she is born. So her eggs are as old as she is, and non-disjunction has more chance to accumulate in them over time. Men produce new sperm daily.

Q.81 [Long] Describe the causes, symptoms and treatment of Down syndrome.

In **Down syndrome**, the total chromosome number is 47 (2n+1). It was first described in 1866 by J. Langdon Down. About 1 in every 750 children has it.

	Down syndrome
Causes	Non-disjunction in autosome pair 21 during the formation of the egg. The egg has 24 chromosomes; when a normal sperm fertilizes it, the child (male or female) has trisomy 21 with 47 chromosomes.
Symptoms	Mental impairment, stunted growth, increased skin on back and neck, flat facial features, small head, short neck and limbs, upward slanting eyes, small ears, poor muscles, protruding tongue and heart defects. Males are usually sterile; females have lower fertility.
Treatment	Proper management improves the quality of life: screening for common problems, medical treatment where indicated, a good family environment, special education and proper care.

Q.82 [Long] Describe the causes, symptoms and treatment of Klinefelter syndrome.

Klinefelter syndrome is the XXY condition (47 chromosomes; 2n+1 or 44+XXY). It was first described by Harry Klinefelter in 1942.

	Klinefelter syndrome
Causes	Non-disjunction of the X chromosomes in the mother gives an XX egg; when a normal Y sperm fertilizes it, the child is XXY. Non-disjunction in the father can give an XY sperm, which with a normal X egg also gives XXY.

	Klinefelter syndrome
Symptoms	Sterile males with many female body characteristics (less body hair, breast growth), small testicles, abnormal body proportions (long legs, short trunk, shoulders equal to hip size); the voice may not be as deep.
Treatment	Testosterone (to compensate its deficiency at puberty); surgical removal of breast; behavioural therapy for language disorders, difficulties at school and socialization.

Q.83 [Short] What is triple X syndrome?

If an XX egg fuses with an X sperm, the female child has 47 chromosomes with XXX sex chromosomes. Such females show learning disabilities, tall stature and wide-spaced eyes. They are fertile and puberty starts at the normal age.

Q.84 [Long] Describe the causes, symptoms and treatment of Turner syndrome.

Turner syndrome is the XO condition in a female (45 chromosomes; 2n–1 or 44+XO), i.e., monosomy in the sex chromosomes. It was first described by Dr. Henry Turner in 1938. It occurs roughly once in every 5000 female births.

	Turner syndrome
Causes	Non-disjunction gives an egg (or sperm) with no sex chromosome (O). If an O egg fuses with a normal X sperm, or an O sperm with a normal X egg, an XO zygote is formed. (An OY zygote cannot develop.)
Symptoms	The female is sterile, with short stature, webbed neck, immature sex organs, low mental abilities, and small chin and jaw. Associated problems: liver and kidney issues, obesity, diabetes and hypertension.
Treatment	Growth hormone treatment started early in childhood can give adult height; estrogen replacement therapy for the development of secondary sexual characteristics.

Q.85 [Short] Why does an OY zygote fail to develop?

Humans cannot survive without the genes present on the X chromosome. An OY zygote has no X chromosome, so it fails to develop further.

Q.86 [Short] Which autosomal trisomies can survive, and for how long?

A zygote with one less autosome cannot complete development, and most with an extra autosome also do not survive. An extra chromosome in pairs 13, 15 or 18 causes severe developmental defects, and such infants die within a few months. Individuals with an extra copy of chromosome 21 or 22 usually survive to adulthood, but they are short, have poor muscles, and their mental development is affected.